Universality of Stochastic Control of Quantum Chaos with Measurement and Feedback

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We investigate universal features of measurement-and-feedback control of quantum chaotic dynamics by examining the quantum Arnold cat map, a paradigmatic model of quantum chaos. Inspired by probabilistic control of classical chaos, our protocol stochastically alternates between intrinsic instability and engineered control operations that steer trajectories toward a target point. Simulation of exact quantum dynamics and a semiclassical truncated Wigner approximation reveal universal properties of the cat map's control transition. To further characterize this universality, we introduce the inverted harmonic oscillator as an analytically tractable effective model of instability. By integrating numerical simulations, a semiclassical Fokker-Planck description, and a direct spectral analysis of the stochastic quantum channel, we identify quantum signatures absent in classical limits. The close agreement between quantum simulation, truncated Wigner approximation, and inverted oscillator analysis shows that universal features of the transition are set by uncertainty-limited quantum fluctuations and are insensitive to genuine quantum interference.

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Introduction—Order emerges from chaotic dynamics across a range of scales, from microscopic quantum systems to macroscopic classical, and even socioeconomic, processes. In many-body systems, local interactions are sufficient to induce ordering. Bird flocks and fish schools provide familiar classical examples [1,2], superconductivity and superfluidity represent analogous quantum phenomena, and within statistical mechanics, ordered states emerge through phase transitions.

Beyond passive emergence, order can be engineered from chaotic dynamics using both deterministic procedures, e.g., the Ott-Grebogi-Yorke method [3], and probabilistic approaches, involving the stochastic intervention of a control map with probability p into chaotic dynamics [4–6]. In this scheme the control map and chaotic dynamics share a periodic orbit, unstable for the latter and stable for the former, which becomes a global attractor above a critical control rate p_c. This is a random multiplicative process where distance from the control point is either amplified by a factor e^{κ} by chaotic dynamics or contracted by a factor $e^{-\gamma}$ by control; this has been studied in turbulent systems [7–10] and market dynamics [11–18]. These ideas have also recently gained resonance in quantum systems, where measurement-conditioned feedback operations can stabilize states and steer dynamics [19–32].

In this Letter, we study stochastic measurement-and-feedback control of quantized chaotic maps, focusing on the quantum Arnold cat map [33]. Using exact quantum dynamics and the truncated-Wigner-approximation (TWA), we extract universal properties of the quantum control transition. Since essential features of the transition are determined by the local dynamics near the control point, it can be effectively described by an inverted harmonic oscillator (IHO) [34]. Focusing on this model, we use numerical simulations of Gaussian-state evolution, a semiclassical Fokker-Planck (FP) analysis augmented by quantum uncertainty, and exact eigenoperators of the stochastic quantum channel, to characterize the transition and the controlled phase.

Despite lacking a compact phase space, genuine quantum chaos, or quantum interference, the IHO shows excellent agreement with controlled cat-map dynamics. Using the cat map as a paradigm, this indicates that for $2D$ chaotic dynamics the essential physics of *quantum* stochastic control is governed locally near the unstable fixed point. The measurement-and-feedback protocol suppresses wave-packet spreading and self-interference, effectively pushing the Ehrenfest time to infinity and rendering interference largely irrelevant for controllability, consistent with the broader role of continuous monitoring in the emergence of classical chaotic dynamics [41].

Classical control—We begin with a brief illustration of classical control of chaos. Take $\mathbf{r}(t)$ to be the small displacement from an unstable fixed point or periodic orbit in a $2D$ phase space, where t is a discrete time index. There is always a choice of coordinates where chaotic and control maps act on $\mathbf{r}(t)$ as $\mathcal{S} = \mathrm{diag}(e^{\kappa}, e^{-\kappa})$ and $\mathcal{C} = \mathrm{diag}(e^{-\gamma}, e^{-\gamma})$, respectively, with $\kappa, \gamma > 0$ parametrizing their respective strengths (det $\mathcal{S} = 1$ due to being area preserving). The behavior of the first component r_1

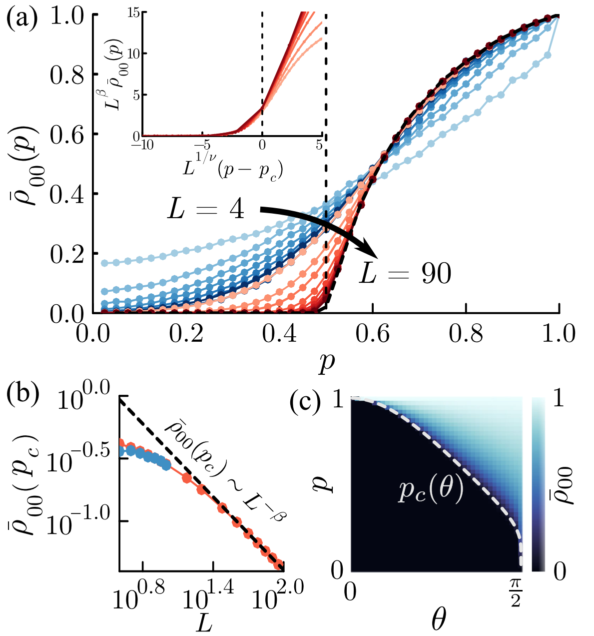

FIG. 1. (a) Order parameter $\bar{\rho}_{00}(p)$ from exact cat-map simulations ($L = \log_2 N = 4$–10, blue) and TWA ($L = 10$–90, red); inset: TWA collapse near $p = p_c$ with $\nu = 1$ and $\beta = 0.97$. (b) $\bar{\rho}_{00}(L)$ at p_c for exact (blue) and TWA (red); dashed line: $\bar{\rho}_{00} \propto L^{-\beta}$ with $\beta = 0.97$. (c) TWA $\bar{\rho}_{00}$ at $L = 90$ vs p, θ, consistent with the phase boundary $p_c(\theta)$ (white dash); $\hbar = (1/2\pi N) = (1/2^{L+1}\pi)$.

determines controllability of the system: chaos is controlled when the Lyapunov exponent $\overline{\ln|r_1(t)|} < 0$ as $t \to \infty$, where $\overline{(\cdots)}$ represents averaging over stochastic trajectories. The average of this quantity can be performed analytically, resulting in a critical control rate [6,42],

$$p_c = \frac{\kappa}{\kappa + \gamma}, \tag{1}$$

above which the system is controlled, and below which chaos prevails.

Quantized chaotic map—We use the Arnold cat map [33] as a paradigmatic example of 2D chaos. Classically, this map transforms the unit square as

$$\mathbf{r}(t+1) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \mathbf{r}(t) \mod 1, \tag{2}$$

and has fixed point $\mathbf{r}_0 = 0$ and strength $\kappa = 2\ln[(1+\sqrt{5})/2]$. To quantize this map we promote the two coordinates to noncommuting quantum operators: $\mathbf{r} = (r_1\, r_2) \to \hat{\mathbf{r}} = (\hat{x}\, \hat{p})$, with $[\hat{x}, \hat{p}] = i\hbar$. The cat map can be implemented as unitary evolution on these operators with $\hat{U}_{\mathrm{cat}} = e^{-i\hat{p}^2/(2\hbar)} e^{i\hat{x}^2/(2\hbar)}$. We take discrete position eigenstates $|x_n = n/N\rangle$ and momentum eigenstates $|p_m = m/N\rangle$, $n, m = 0, 1, \ldots, N-1$. With $|x_n\rangle = (1/N)\sum_m e^{2\pi i n m/N}|p_m\rangle$ and unit-square phase space, $\hbar = (1/2\pi N)$ [43,44]. The continuum limit $N \to \infty$ is therefore the classical limit.

Because unitary quantizations arise from area-preserving maps [45,46], implementing non-area-preserving control requires measurement. We do this with a positive operator-valued measure (POVM) defined by Kraus operators. To construct them, we introduce a basis via $[2 - \cos(2\pi\hat{x}) - \cos(2\pi\hat{p})]|n_c\rangle = \epsilon_{n_c}|n_c\rangle$ such that $|0\rangle$ is the controlled state (fixing the phases of the basis by the harmonic oscillator convention, see Supplemental Material [43]). This allows us to construct Kraus operators

$$\hat{K}_m(\theta) = \frac{(-i)^m}{\sqrt{m!}} \sin^m(\theta)(\cos(\theta))^{\hat{a}^\dagger\hat{a}}\hat{a}^m, \tag{3}$$

where $\theta \in [0, \pi/2]$ parametrizes the strength of control, $\hat{a} = \sum_{n_c=0}^{N_{\mathrm{ctrl}}-1} \sqrt{n_c}|n_c - 1\rangle\langle n_c|$ where we take $N_{\mathrm{ctrl}} = N/2$, and complete the POVM with a projector on the remaining basis states. This control channel acts on $\hat{x}$ and $\hat{p}$ as a rescaling by $\cos\theta$, so classical and quantum control strengths γ and θ are related as $\cos\theta = e^{-\gamma}$.

Both quantum channels (cat map and control) can be simulated with the TWA [47]. In the TWA, dynamics are classical with quantum fluctuations sampled from the initial Wigner function $W_0(x, p) = (1/\pi\hbar)e^{-(x^2+p^2)/\hbar}$, while control induces noise $(x, p) \mapsto \cos\theta(x, p) + \boldsymbol{\eta}$ with $\boldsymbol{\eta}$ Gaussian of variance $(\hbar/2)(1 - e^{-2\gamma})$ (see Supplemental Material [43]). We use the late-time, trajectory-averaged squared overlap with the control state, as an order parameter for control, $\bar{\rho}_{00} = \overline{\langle 0|\hat{\rho}(t \to \infty)|0\rangle}$ where $\hat{\rho}(t)$ is the density matrix, which we can compute via the TWA under the full channel. This formulation neglects interference and follows from the linear approximation in Eq. (10), so control-POVM nonlinearities are neglected as well.

In Fig. 1(a) we show $\bar{\rho}_{00}(p)$ for finite $\hbar$, where the transition is *rounded* into a crossover but sharpens as $\hbar \to 0$ (i.e., $N \to \infty$). Furthermore, we see this crossover smoothly connect the quantum data for $L \equiv \log_2 N = 4$–10 to the TWA for $L = 10$–90; the two agree quantitatively for $L = 10$ ($N = 1024$) [43]. The control strength θ is chosen so that $p_c = 1/2$ as calculated by Eq. (1). Scaling collapse near $p = p_c$ gives correlation length exponent $\nu \approx 1$, and examining $\bar{\rho}_{00}$ as a function of L at p_c, shown in Fig. 1(b), gives the order parameter scaling exponent $\beta \approx 1$. Figure 1(c) shows $\bar{\rho}_{00}$ obtained with the TWA for $L = 90$ for all p and θ, and clearly identifies the boundary between controlled and uncontrolled phases, agreeing with the expected classical result for p_c.

The success of the TWA in capturing the behavior of the quantum cat map's control transition indicates that quantum interference effects are largely irrelevant, and suggests that the transition admits a universal description. Motivated by this, we now isolate the dynamics near the unstable fixed point.

Inverted harmonic oscillator—The saddle-point structure near an unstable fixed point is modeled by the IHO,

$$\hat{H} = \frac{\hat{p}^2}{2} - \frac{\Omega^2 \hat{x}^2}{2}, \tag{4}$$

where Ω controls the strength of the potential. Time evolution under this Hamiltonian $\hat{U}(t) = \exp(-i\hat{H}t/\hbar)$ generates a single-mode squeezing operation and a Heisenberg-picture quantum channel

$$\mathbb{S}[\hat{\mathcal{O}}] = \hat{U}^\dagger(t)\hat{\mathcal{O}}\,\hat{U}(t). \tag{5}$$

The action of this channel for time t_S models the effect of a quantum-chaotic map near the fixed point with strength $\kappa = \Omega t_S$. The operators $\hat{v}_\pm \equiv \sqrt{\Omega/2}(\hat{x} \pm \hat{p}/\Omega)$ grow or decay as $\mathbb{S}[\hat{v}_\pm] = e^{\pm\kappa}\hat{v}_\pm$, and all operators $(\hat{v}_+)^m(\hat{v}_-)^n$ are eigenvectors of the channel; $\langle \hat{v}_\pm \rangle$ are analogs of the components of the classical $\mathbf{r}$ above.

As for the quantum cat map we can define ladder operators $\hat{a}$, $\hat{a}^\dagger$ and number states $|n\rangle$. The vacuum $|0\rangle$ is the unstable fixed point of IHO evolution. Control of the IHO can be implemented by introducing an ancilla mode with its own ladder operators $\hat{b}, \hat{b}^\dagger$ and number states $|n\rangle_b$. Initializing the ancilla in its vacuum $|0\rangle_b$, coupling it to the IHO, then measuring and resetting it back into the vacuum state yields Kraus operators $K_m(\theta) = \langle m|_b e^{-i\theta(\hat{a}^\dagger\hat{b} + \hat{a}\hat{b}^\dagger)}|0\rangle_b$, which have exactly the same form as Eq. (3). The Heisenberg-picture quantum channel for control is

$$\mathbb{C}[\hat{\mathcal{O}}] = \sum_{m=0}^{\infty} \hat{K}_m^\dagger(\theta)\hat{\mathcal{O}}\hat{K}_m(\theta). \tag{6}$$

All normally ordered products of $\hat{a}$ and $\hat{a}^\dagger$ are eigenvectors of this channel, $\mathbb{C}[(\hat{a}^\dagger)^k\hat{a}^l] = \cos^{k+l}(\theta)(\hat{a}^\dagger)^k\hat{a}^l$. As for Eq. (3), $\mathbb{C}$ scales linear operators by $\cos\theta = e^{-\gamma}$.

The full stochastic evolution of a generic operator acting on the IHO is the weighted sum of these two channels

$$\hat{\mathcal{O}}(t+1) = \mathbb{T}[\hat{\mathcal{O}}(t)] = (1-p)\mathbb{S}_\kappa[\hat{\mathcal{O}}(t)] + p\mathbb{C}_\gamma[\hat{\mathcal{O}}(t)], \tag{7}$$

where channels are labeled by their respective exponential parameters.

Quantum dynamics and Gaussian state evolution—The dynamics of $\langle \hat{v}_\pm \rangle$ under $\mathbb{S}_\kappa$ and $\mathbb{C}_\gamma$ follow the classical dynamics, and quantum effects manifest only at higher orders in canonical operators. Gaussian states are characterized by their mean $\langle \hat{\mathbf{r}} \rangle$ and covariance matrix,

$$\sigma_{ij} = \frac{1}{2}\langle\{\hat{r}_i, \hat{r}_j\}\rangle - \langle \hat{r}_i\rangle\langle \hat{r}_j\rangle \tag{8}$$

where $\hat{\mathbf{r}} = (\hat{v}_+\,\hat{v}_-)^T$ and $\{\cdot,\cdot\}$ is the anticommutator. To highlight the quantum aspects of control we take $\langle \hat{\mathbf{r}} \rangle = 0$ so the system is classically controlled. Because both $\mathbb{S}_\kappa$ and $\mathbb{C}_\gamma$ are Gaussian channels, Gaussian states remain Gaussian and we only need to track evolution of σ,

$$\mathbb{S}_\kappa[\sigma] = \begin{pmatrix} e^{2\kappa}\sigma_{11} & \sigma_{12} \\ \sigma_{12} & e^{-2\kappa}\sigma_{22} \end{pmatrix}, \tag{9}$$

$$\mathbb{C}_\gamma[\sigma] = e^{-2\gamma}\sigma + \frac{\hbar}{2}(1 - e^{-2\gamma})\mathbf{1}. \tag{10}$$

By design $\mathbb{S}_\kappa$ is a single-mode squeezing channel, and $\mathbb{C}_\gamma$ is a pure-loss attenuator [48]. For $p \neq 0$ the off-diagonal elements of σ vanish for $t \to \infty$ due to control, so $\sigma_{11}, \sigma_{22} \equiv \sigma_\pm$ alone determine steady state properties. These are the variances of canonical operators, so the constant term in $\mathbb{C}_\gamma$ manifests the quantum uncertainty relation between them, $\sigma_+\sigma_- \geq \hbar^2/4$, and the control point is $\sigma_\pm = \hbar/2$. We note now that Eq. (10) is the basis for the noise inserted via the TWA control protocol mentioned earlier. That noise is the classical representation of the uncertainty-limited broadening imposed by the quantum control channel. The actions of $\mathbb{S}_\kappa$ and $\mathbb{C}_\gamma$ on $\sigma_\pm$ do not commute, complicating the analysis of trajectories of $\sigma_\pm$, however the system can be analyzed numerically together with analytical methods in applicable limits described later. For convenience we scale $\sigma_\pm$ by $\hbar$ in our further analysis, leaving them dimensionless.

Semiclassical analysis of quantum control transition—As above we use $\bar{\rho}_{00}$ as the order parameter, with the controlled state signaled by $\bar{\rho}_{00} > 0$. With our assumption $\langle \hat{\mathbf{r}} \rangle = 0$, for a Gaussian trajectory $\rho_{00}(t) = [(\sigma_+(t)/+\frac{1}{2})(\sigma_-(t)+\frac{1}{2})]^{-1/2}$ [48]. Since an arbitrary density matrix can be written as a linear combination of Gaussian density matrices, the order parameter $\bar{\rho}_{00}$ for a generic state is

$$\bar{\rho}_{00} = \int d\sigma_+ d\sigma_- \frac{\mathcal{Q}(\sigma_+, \sigma_-)}{\sqrt{\left(\sigma_+ + \frac{1}{2}\right)\left(\sigma_- + \frac{1}{2}\right)}}, \tag{11}$$

where $\mathcal{Q}(\sigma_+, \sigma_-)$ is the steady-state probability density over Gaussian trajectories, which obeys

$$\mathcal{Q}(\sigma_+, \sigma_-) = (1-p)\mathcal{Q}(\mathbb{S}_\kappa^{-1}[\sigma_+, \sigma_-]) + p\mathcal{Q}(\mathbb{C}_\gamma^{-1}[\sigma_+, \sigma_-]). \tag{12}$$

This is analogous to the Frobenius-Perron equation for the invariant density used in classical chaotic dynamics [49,50]. We can compute Eq. (11) numerically.

In Fig. 2 we show results computed with the IHO corresponding to those presented for the cat map in Fig. 1. The order parameter's dependence on p shown in Fig. 2(a)

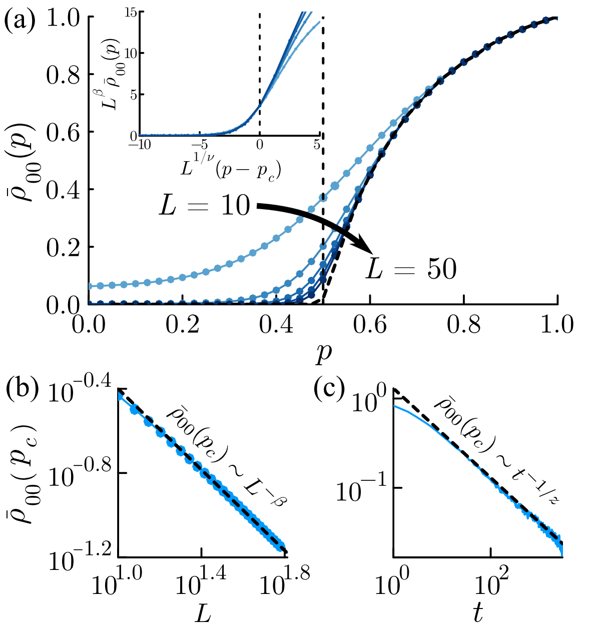

FIG. 2. (a) Steady-state $\bar{\rho}_{00}(p)$ from the fixed point of $\mathbb{T}$ for $\kappa = \gamma = 2\ln[(1+\sqrt{5})/2] \approx 0.42$, with cutoffs $2^{-L-1} < \sigma_{\pm} < 2^{L-1}$ for $L = 10, 20, 30, 40, 50$; dashed: late-time average over 5000 trajectories; inset: collapse near $p = p_c$ with $\nu = 1$ and $\beta = 0.96$. (b) $\bar{\rho}_{00}(L)$ at p_c (same cutoffs); dashed: $\bar{\rho}_{00} \propto L^{-\beta}$ with $\beta = 0.96$. (c) $\bar{\rho}_{00}(t)$ at p_c; dashed: $\bar{\rho}_{00} \propto t^{-1/z}$ with $z = 2$.

is computed both as an average over stochastic trajectories and as the fixed-point of the quantum channel after imposing cutoffs $\sigma_+ \in (1/2, 2^{L-1})$ and $\sigma_- \in (2^{-L-1}, 1/2)$ [51]. The two methods agree for large L, with quick convergence for $p > p_c$ since values of $\sigma_{\pm}$ far from the control point are rarely explored. Despite considering only classically controlled states $\langle \hat{\mathbf{r}} \rangle = 0$ we see a clear control transition sharpening at p_c for large L. Critical properties at p_c are obtained from L and t dependence of $\bar{\rho}_{00}$, shown in Figs. 2(b) and 2(c), and by collapsing the p dependence of $\bar{\rho}_{00}$ near $p = p_c$. We find $\beta \approx 1$ and $\nu \approx 1$, agreeing with the result of the cat map, and also dynamical exponent $z \approx 2$, all characteristic of random walk universality.

Fokker-Planck analysis—Since $\sigma_- \in (0, 1/2)$ at late times, $\bar{\rho}_{00}$ vanishing or remaining nonzero must result from σ_+ diverging for $p < p_c$ and remaining finite for $p > p_c$ in Eq. (11). The stochastic dynamics of σ_+ are a random multiplicative process with lower bound $\sigma_+ > 1/2$ [52]: $\sigma_+(t+1) = \Lambda_t \sigma_+(t)$ where Λ_t is drawn randomly from $(e^{2\kappa}, e^{-2\gamma})$ with corresponding probabilities $(1-p, p)$, and $\sigma_+ > 1/2$ imposed by hand. In the limit of weak dynamics (κ, γ small) and defining $y = \ln \sigma_+$, we derive a Fokker-Planck (FP) equation for $P(y, t)$, the distribution of values taken by y during these dynamics,

$$\frac{\partial P(y,t)}{\partial t} = -\bar{v}\frac{\partial P(y,t)}{\partial y} + \mathcal{D}\frac{\partial P(y,t)}{\partial y}, \tag{13}$$

with drift velocity $\bar{v} = 2\kappa(p_c - p)/p_c$ and diffusion constant $\mathcal{D} = 2\kappa^2 p(1-p)/p_c^2$. This can be solved for both $P(y, t)$ far from the boundary $y^{\min} = -\ln 2$ and for the nontrivial steady state $P(y)$ for $p > p_c$,

$$P(y,t) \sim \frac{1}{\sqrt{2\pi\mathcal{D}t}} e^{-\frac{(y-\bar{v}t)^2}{2\mathcal{D}t}}, \qquad y \gg y^{\min}, \tag{14}$$

$$P(y) = \frac{\xi}{2^{\xi}} e^{-\xi y} \Rightarrow \mathcal{Q}_+(\sigma_+) = \frac{\xi}{2^{\xi}} \sigma_+^{-1-\xi}, \qquad p > p_c, \tag{15}$$

where $\xi = |\bar{v}|/\mathcal{D}$. The distributions for σ_+ are obtained as $\mathcal{Q}_+(\sigma_+) = P(\ln \sigma_+)/\sigma_+$, and have log-normal and power-law forms, respectively [53]. In Supplemental Material [43] we present the details of the FP derivation and demonstrate this log-normal behavior of $\mathcal{Q}_+(\sigma_+, t)$.

With solutions for the distribution we can examine criticality of the transition. At $p = p_c$, $\bar{v} = 0$ so the distribution has no overall drift and y evolves diffusively away from the boundary, yielding a divergence in y (and σ_+) at late times,

$$\bar{\rho}_{00}(t) \sim \int_{\frac{1}{2}}^{\infty} \mathrm{d}\sigma_+ \frac{\mathcal{Q}_+(\sigma_+, t)}{\sqrt{\sigma_+(t)}} \sim \int_0^{\infty} \mathrm{d}y \frac{e^{-\frac{y^2}{2\mathcal{D}t}}}{\sqrt{\mathcal{D}t}} e^{-\frac{y}{2}} \sim t^{-1/2}. \tag{16}$$

If we also impose a maximum $\sigma_+^{\max} = 2^L$, giving $y^{\max} = L\ln 2$, then for late times and large L the hard wall boundaries combined with the diffusive nature of P give $P(y) \sim 1/L$ and we have

$$\bar{\rho}_{00}(L) \sim \int_{\frac{1}{2}}^{2^L} \mathrm{d}\sigma_+ \frac{\mathcal{Q}_+(\sigma_+)}{\sqrt{\sigma_+}} \sim \int_0^{L\ln 2} \mathrm{d}y \frac{e^{-\frac{y}{2}}}{L} \sim \frac{1}{L}. \tag{17}$$

The Fokker-Planck analysis therefore recovers the critical behavior observed numerically in Figs. 2(b) and 2(c) with exponents $\beta = 1$ and $z = 2$.

The multiplicative dynamics used to obtain Eq. (13) are approximate. The map $\mathbb{C}_\gamma$ imposes the constraint $\sigma_+ > 1/2$ in a smooth way, while the analysis here imposed it crudely, with the difference largest for $\sigma_+ \sim 1/2$. However, the universal results Eqs. (16) and (17) depend only on the large-σ_+ properties of Eq. (13)—diffusive behavior away from the boundary and normalizability of distributions—and critical properties are insensitive to precise behavior near $\sigma_+ \sim 1/2$. This is demonstrated in Supplemental Material [43] by derivation of the FP equation from exact $\mathbb{S}_\kappa$ and $\mathbb{C}_\gamma$.

Exact quantum dynamics—The FP analysis developed above is semiclassical, but exact quantum results can be obtained as well. Since $\hat{\mathbf{1}}$ and $\hat{v}_+$ are eigenvectors of $\mathbb{T}$ with eigenvalues 1 and $(1-p)e^{\kappa} + pe^{-\gamma}$, an inductive argument can be developed using these base cases to give an infinite set of eigenvectors

$$\hat{Z}_n = (\hat{v}_+)^n + \sum_{k=1}^{\lfloor n/2 \rfloor} \alpha_{n,k} \hat{Z}_{n-2k} = \sum_{k=0}^{\lfloor n/2 \rfloor} \beta_{n,k} (\hat{v}_+)^{n-2k}, \quad (18)$$

with eigenvalues $\lambda_n = (1-p)e^{n\kappa} + pe^{-n\gamma}$, i.e., $\mathbb{T}[\hat{Z}_n] = \lambda_n \hat{Z}_n$, where the $\lfloor n/2 \rfloor$ coefficients $\alpha_{n,k}$ or $\beta_{n,k}$ are found by solving a system of linear equations. The full construction is detailed in Supplemental Material [43].

When $\lambda_n < 1$ the operator $\hat{Z}_n$ is controlled, $\langle \hat{Z}_n \rangle \to 0$ as $t \to \infty$. Since the eigenvalues of all other order-n eigenoperators are less than one at $p = p_c$, control of $\hat{Z}_n$ diagnoses controllability for all operators of order n. We thus obtain

$$p_n^* = \frac{e^{n\kappa} - 1}{e^{n\kappa} - e^{-n\gamma}} > p_c \quad \forall\ n > 0 \quad (19)$$

as the control probability for all order-n operators, with $\lim_{n\to 0} p_n^* = p_c$. Since $p_{n+1}^* > p_n^* > p_c$ for $n > 0$, as p increases above p_c there is a sequence of p values where operators of increasing order become finite. In the End Matter, we relate the sequence p_n^* to moments of the steady-state distribution obtained from our Fokker-Planck analysis becoming finite.

Conclusion—We elucidate universal features of stochastic control transitions in quantum chaotic systems exemplified by the quantum cat map. Agreement between exact dynamics, TWA, and the controlled IHO shows that these universal features are set by uncertainty-limited fluctuations and are largely insensitive to quantum interference. This framework explains the emergence of random-walk universality and how quantum fluctuations dress the controlled phase. Previous work [19,28,30,54,55] demonstrated stochastic quantum control *without* full canonical quantization and hence lacked a formal connection to quantum chaos. In this Letter, uncertainty-limited localization and the operator-moment hierarchy provide a genuinely quantum signature; this is underscored by the fact that the small-$\hbar$ physics is simulable with a classical control map with quantum-limited noise [43]. Nonetheless, due to the power-law nature of the steady-state Eq. (15), there is potential for interference phenomenon to be seen in the nature of higher moments beyond the analysis here; we leave that analysis to future work.

A direct platform for the IHO is a driven Kerr/parametric oscillator in cavity or circuit QED; above threshold the origin (vacuum) is an unstable fixed point and the dynamics form an effective phase-space double well while measurement-based feedback or autonomous reservoir engineering implements the attenuation control channel [56–59]. Collective-spin protocols also realize nonlinear dynamics and monitored entanglement regimes [60,61]. Fully chaotic maps with clear unstable orbits have cold atomic candidates, where analogous control protocol could be designed [62,63].

As noted above, the generic theory of the IHO cannot access the uncontrolled phase of specific maps. Other quantum phenomena within that phase, such as qubit encoding, interference, and entanglement, can occur and are rich areas for future development. Beyond these, a further question remains of how the *many-body* problem behaves, where the phase space is extensive and Lyapunov exponents can undergo a transition themselves.

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Data availability—The data that support the findings of this article are openly available [65].

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End Matter

Though our main focus has been on the control transition itself, we can examine generic properties of the controlled phase using both distributions obtained from the Fokker-Planck analysis and exact quantum channel results of the IHO.

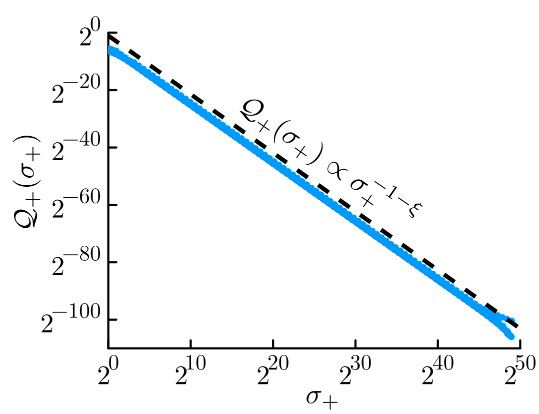

FIG. 3. Comparison of the power-law form of the distribution $\mathcal{Q}_+(\sigma_+)$ acquired from the FP analysis with the distribution of σ_+ computed numerically from the fixed point of $\mathbb{T}$ with $\kappa = \gamma = 0.2$ and cutoff $L = 50$ for $p = 0.6$.

In Fig. 3, we compare the power-law form of $\mathcal{Q}_+(\sigma_+)$ in Eq. (15) to the distribution computed numerically for $p > p_c$. The lower boundary of the FP dynamics is vital for the existence of a late-time steady state in the controlled phase—in Eq. (13) $\bar{v} < 0$ for $p > p_c$, so the distribution drifts to small y and is held up by the barrier at $y^{\min}$. These two opposing forces balance to yield a stable distribution.

We can further compute moments of this distribution

$$\overline{\sigma_+^n} = \int d\sigma_+ \sigma_+^n \mathcal{Q}_+(\sigma_+) = \frac{2^{-\xi}\xi}{\xi - n}, \tag{A1}$$

and find that the nth moment of σ_+ is finite only when $\xi > n$. Setting $\xi = n$ gives a sequence of control rates at which these moments become finite in the FP framework,

$$p_{2n}^{\mathrm{FP}} = p_c + (1 - p_c)n\kappa + O(\kappa^2) > p_c, \tag{A2}$$

where we have expanded the result in small κ since this was necessary to obtain Eq. (13) and $\mathcal{Q}_+(\sigma_+)$. The nth moment of $\mathcal{Q}_+$ is related to quantum operators of order $2n$ since σ_+ is quadratic.

This sequence of moment control rates are directly related to the sequence of control rates p^* in Eq. (19) for which quantum operators of order n become controlled: $p^*_{2n} = p^{\mathrm{FP}}_{2n}$ to first order in κ. This confirms that the FP analysis captures key features of the exact quantum dynamics in the appropriate limits, and equivalently demonstrates that the operators $\hat{Z}_n$ generalize the moment-control picture of the semiclassical analysis to a quantum context.